Ainesh Sabharwal

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Education

University of California, Berkeley — Berkeley, CA
Bachelor of Arts in Computer Science — *Aug. 2026 – May 2029*

Bridgeland High School — Cypress, TX
High School Diploma, Graduated Salutatorian (2/921 people) — *Aug. 2022 – May 2026*

Experience

Undergraduate Research Assistant — Sept. 2026 – Present
University of California, Berkeley — *Berkeley, CA*

- Building a Gemini extraction pipeline with structured JSON outputs to classify municipal regulations from LOCUS's 2.2M-row corpus
- Developing Python and DuckDB filters to recover 168K overlooked licensing and permitting sections
- Creating a multi-annotator review interface to validate model outputs against human annotations

Lead Instructor — Oct. 2025 – Mar. 2026
iCode School Franchise — *Cypress, TX*

- Taught Python, Java, and Roblox development fundamentals through hands-on projects for K–12 students
- Partnered with the lead director to redesign standardized curriculum to match students' skill levels
- Guided students through coding projects, building debugging and problem-solving skills

Unofficial Teaching Assistant — Aug 2025 – May 2026
Self-Employed / Bridgeland High School — *Cypress, TX*

- Volunteered with the CS department to support grading for a high school Data Structures and Algorithms course
- Created instructional videos to explain algorithms and reinforce programming concepts
- Developed worksheets and practice exercises to strengthen students' understanding of data structures and problem-solving

Projects

github.com/ainesh-sabharwal

Age Estimator | *Python, YOLOv8, OpenCV, ONNX, PyTorch* — Oct. 2026 – Present

- Developed a multi-model computer vision pipeline to estimate age distributions from images and video, combining face detection, 6-view test-time augmentation, and body-based classification
- Trained and evaluated models on 100K+ images from UTKFace, FairFace, APPA-REAL, and PA-100K, reducing 16–17 age-band MAE from 6.85 to 4.01
- Achieved 84.5% minor recall with a 1.3% false-positive rate and designed uncertainty-aware predictions to identify cases where visual age estimation is unreliable

ApathyAI | *Python, Tauri, Ollama, LanceDB* — June 2026 – Aug. 2026

- Co-developed a desktop AI copilot that interprets screen context and summarizes documents using local LLMs
- Integrated accessibility APIs, OCR, and vector search to retrieve relevant context from recent workflows
- Added API authentication, rate limiting, and sensitive-data redaction to protect captured screen content

Bridgeland Arena Hub | *TypeScript, React, Supabase, React Three Fiber* — Oct. 2024 – July 2025

- Co-developed an event ticketing platform that placed 3rd nationally in FBLA Website Design
- Built interactive 2D and 3D seating charts with seat selection and section-based pricing
- Integrated PayPal checkout, ticket history, and administrative tools for event management

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, Swift, HTML/CSS
Frameworks: React, SwiftUI, Tauri, Tailwind CSS
Developer Tools: Git, GitHub, PostgreSQL, Supabase, DuckDB, Vite, Vercel
AI/ML & APIs: PyTorch, OpenCV, YOLOv8, Gemini API, Ollama, LanceDB